



HERDER
ELEKTRONISCHE SYSTEMEN



○ EMBEDDED SYSTEMS, CONTROLS AND TEST DEVELOPMENT

Hardware, firmware, *made to work as one.*

Herder Elektronische Systemen helps recover, build and validate electronic systems across hardware, firmware, controls and instrumentation.

DISCUSS YOUR PROJECT →

EXPLORE OUR EXPERTISE ↓



SERVING CLIENTS INTERNATIONALLY EMBEDDED SYSTEMS, CONTROLS AND ELECTRONICS
SCROLL TO EXPLORE ↓

Engineering status
IN REVIEW

Engineering across the interfaces *where prototypes usually fail.*

The difficult part is often the space between disciplines. We bring hardware, firmware and control together, then test the system as a whole.

01



Embedded controls and firmware

We design firmware around the hardware it has to control, then document the interfaces, states and tests needed to move the prototype forward.



02



Power electronics prototyping

We help shape converter prototypes, bring them up on the bench and use measurements to guide the next design decision.



03



Test automation and validation

We build repeatable tests that connect instruments, data and analysis, so the results are easier to trust and review.

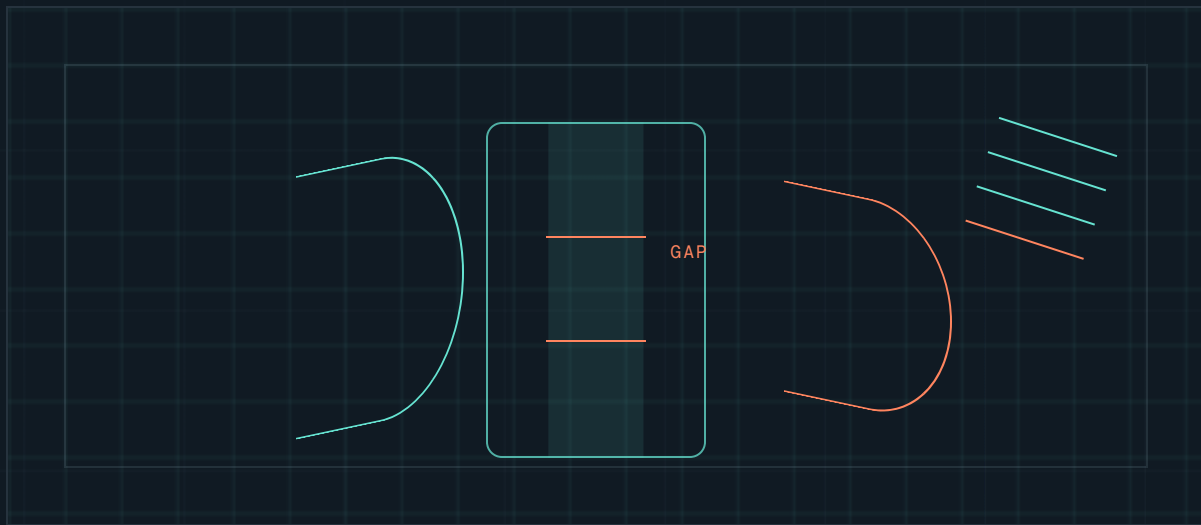
04



System integration and technical recovery

When a project has inherited code or unclear behavior, we trace the problem across hardware, firmware and communications and leave a documented path forward.

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02 ——— SPECIALIST MAGNETICS

CUSTOM MAGNETICS / SMPS DEVELOPMENT

Custom high- frequency *transformer prototypes.*

Transformer definition, winding construction, 3D modeling and bench evaluation for switch-mode power-supply development.

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02 — OUR APPROACH

Make the critical decision *while it is still cheap to change.*

We use requirements, prototypes and measurements to expose the important tradeoffs before they become expensive failures.

BRING US YOUR CHALLENGE →

A

Understand

We start with the system, its users and its constraints.

B

Engineer

We test the assumptions that matter, integrate the pieces and resolve the problems that block progress.

C

Enable

You leave with a working system, the evidence behind it and documentation another engineer can use.

03 ——— SELECTED WORK

Representative *technical projects.*

These projects show how the work looks in practice: working hardware, firmware, controls and validation. They are public technical projects, not completed client contracts.

EMBEDDED CONTROLS

01

ESP32 Robotics Control Platform

A complete robotics-control platform integrating inertial sensing, wireless operator input, encoder feedback, motor drive, servo control and self-balancing PID behavior.

ESP32 / C++

IMU / PID

Encoders

MATLAB / Simulink

[VIEW PROJECT](#) →

[VIEW REPOSITORY](#) ↗

POWER ELECTRONICS

02

5V / 1A SMPS

An offline flyback prototype documented from transformer winding and component selection through waveform capture and bench validation.

[Flyback SMPS](#)[TNY285](#)[Waveforms](#)[Validation](#)[VIEW PROJECT](#) →[VIEW REPOSITORY](#) ↗

SAFETY-ORIENTED CONTROLS

03

TI C2000 Actuator Controller

A hardware-backed F28069M demonstrator for command handling, explicit state control, PWM output, setpoint validation and fault response.

[TI C2000](#)[SCI / UART](#)[ePWM](#)[Python tools](#)[VIEW PROJECT](#) →[VIEW REPOSITORY](#) ↗

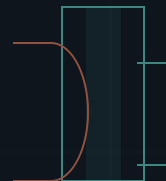
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SPECIALIST MAGNETICS

04

Custom high-frequency transformer prototypes

Transformer definition, winding construction, 3D modeling and measured bench evaluation for SMPS development.

[Core geometry](#)[Custom winding](#)[3D models](#)[Bench evaluation](#)[EXPLORE CUSTOM MAGNETICS](#) →

Controls at *testbed scale.*

PUBLIC UNIVERSITY RESEARCH AND ENGINEERING WORK

TI Piccolo dual-motor PMSM dynamometer

A dual-motor dynamometer built to develop, exercise and validate PMSM control across embedded firmware, CAN communications, real-time testing and measured torque-speed data.

Jonathan's contribution

Jonathan integrated the dual-motor control system, migrated telemetry from SCI to CAN, connected the testbed to Speedgoat real-time hardware and developed validation workflows for speed, torque, power and efficiency data.

Evidence and methods

The repository includes architecture documentation, hardware integration notes, MATLAB/Python processing, efficiency maps and measured speed, torque and power data.

Publication boundary

This work was completed in a university laboratory setting. It is not a private client engagement or an exclusively owned product. The site links to the public repository without reproducing its models or large source files.

[VIEW PROJECT](#) → [VIEW REPOSITORY ↗](#)

05 ——— TECHNICAL RANGE

Built for the *whole picture.*

01 **System architecture**
From requirements to clear interfaces

02 **Analog & digital electronics**
Design support for prototypes

03 **Industrial connectivity**
Communication that fits the system

04 **Verification & validation**
Results tied to test conditions

Architecture → Prototype → Integration → Validation

The process keeps risk visible and gives each phase a clear next step.

Engineering with *direct ownership.*

Herder Elektronische Systemen is an independent engineering practice led by Jonathan, focused on embedded systems, controls, electronics, test development and system integration.

The practice is intentionally small and hands-on, rather than a large staff or production organization.

Let's make it *real.*

Send a short brief. We'll ask the right questions, clarify the next step and help define what the work needs.

- Project type and current stage
- Technical problem, target schedule and expected deliverables
- Approximate budget range and preferred contact method
- If files are needed, we will arrange secure exchange after the initial discussion

Project details are handled confidentially. Please do not send sensitive files in the initial email; secure file exchange can be arranged afterward. Public portfolio material is published only when authorized.

START A TECHNICAL DISCUSSION →

Or write directly: info@herdersystemen.com



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